

Section 3.3 Derivatives of Trig Functions

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\sec x)' = \sec x \cdot \tan x$$

$$(\csc x)' = -\csc x \cdot \cot x$$

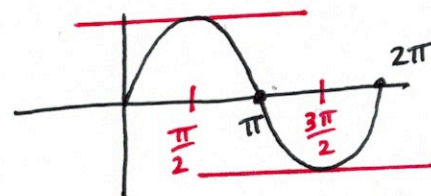
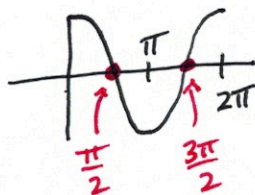
Ex ① $y = \sin x$ on $[0, 2\pi]$

where is $y' = 0$?

$$y' = \cos x$$

$$0 = \cos x$$

$$x = \frac{\pi}{2}$$
$$x = \frac{3\pi}{2}$$



Ex ② $y = x^2 \cdot \sec x$

$$y' = 2x \cdot \sec x + x^2 \cdot \sec x \cdot \tan x$$

$$y' = x \cdot \sec x (2 + x \tan x)$$

Ex ③ $y = \frac{\tan x}{x}$

$$y' = \frac{x \cdot \sec^2(x) - \tan x \cdot 1}{x^2}$$

Ex ④ $f(x) = e^x \cdot \csc x$

$$f'(x) = e^x \cdot \csc x + e^x \cdot (-\csc x \cdot \cot x)$$

$$f'(x) = e^x \csc x (1 - \cot x)$$

Ex ⑤ Find equation of tangent line
for $f(x) = x \cdot \cos x$ when $x = 0$

$$y = 0 \cdot \cos 0 = 0$$

$$f'(x) = 1 \cdot \cos x + x(-\sin x)$$

$$f'(0) = \cos(0) + 0(-\sin 0)$$

$$\text{slope} = 1$$

$$y - 0 = 1 \cdot (x - 0)$$

$$\boxed{y = x} \checkmark$$

Ex ⑥ Can $y' = 0$?

$$y = \cot x$$

$$y' = -\csc^2 x$$

$$(-\sin^2 x) 0 = -\frac{1}{\sin^2 x} (-\sin^2 x)$$

$$0 = 1 \text{ False!}$$

y' cannot equal zero!

Ex ⑦ Prove $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$

$$\frac{d}{dx}(\sec x) = \frac{d}{dx} \left(\frac{1}{\cos x} \right)$$

$$= \frac{0 \cdot \cos x - 1 \cdot (-\sin x)}{\cos^2 x}$$

$$= \frac{\sin x}{\cos^2 x} = \frac{\boxed{1} \cdot \boxed{\sin x}}{\boxed{\cos x} \cdot \boxed{\cos x}}$$

$$\begin{array}{c} \downarrow \qquad \downarrow \\ = \sec x \cdot \tan x \quad \checkmark \end{array}$$